

We show when the polynomial congruence is a field.

Let k be a field and $0 \neq g(x) \in k[x]$.

Lemma 1. *Let $f(x) \in k[x]$. Then $[f(x)] = [0]$ iff $f(x) \in (g(x))$.*

Proof. By the division algorithm, $f(x) = g(x)q(x) + r(x)$ for some $q(x), r(x) \in k[x]$ where $r(x) = 0$ or $\deg r(x) < \deg g(x)$. Since $g(x)q(x) \in (g(x))$, we get $f(x) \in (g(x))$ iff $r(x) \in (g(x))$. When $r(x) = 0$, it follows $r(x) \in (g(x))$. Suppose $r(x) \neq 0$. Then $\deg r(x) < \deg g(x)$ implies $r(x) \notin (g(x))$. Therefore we have $f(x) \in (g(x))$ iff $r(x) \in (g(x))$ iff $r(x) = 0$ iff $f(x) = [0]$. $\square$

Proposition 1. *Then $k[x] \bmod g(x)$ is a field iff $g(x)$ is irreducible in $k[x]$.*

Proof. ($\implies$) Suppose $k[x] \bmod g(x)$ is a field. Then in particular it is a domain. We claim $(g(x))$ must be a prime ideal. Indeed, suppose not, then there exists $a(x), b(x) \notin (g(x))$ such that $a(x)b(x) \in (g(x))$. By Lemma 1, $[a(x)]$ and $[b(x)]$ are nonzero in $k[x] \bmod g(x)$. In particular, $[a(x)][b(x)] = [g(x)] = [0]$. This implies $[a(x)]$ and $[b(x)]$ are zero-divisors and $k[x] \bmod g(x)$ is not a domain. Recall $(g(x))$ is prime ideal iff $g(x)$ is a prime element in $k[x]$. But $k[x]$ is a UFD where prime elements are precisely the irreducibles elements. Hence $g(x)$ must be irreducible.

($\impliedby$) Suppose $g(x)$ is irreducible. Recall $k[x]$ is a PID. Then $(g(x))$ is a maximal ideal in $k[x]$. Let $[f(x)] \neq [0]$ in $k[x] \bmod g(x)$. By Lemma 1, $f(x) \notin (g(x))$. This implies $(g(x), f(x))$ is an ideal that strictly contains the maximal ideal $(g(x))$. Hence $(g(x), f(x)) = k[x]$ and in particular there exists $a(x)$ and $b(x) \in k[x]$ such that $a(x)g(x) + b(x)f(x) = 1$. This shows $[b(x)][f(x)] = [1]$ in $k[x] \bmod g(x)$ and $[f(x)]$ has an inverse. $\square$

Remark. Summarizing the proof above, one can also note that $k[x] \bmod g(x)$ is a domain iff $(g(x))$ is a prime ideal iff $g(x)$ is a prime element in $k[x]$. But since $k[x]$ is a UFD where prime elements agrees with the irreducibles elements. Therefore $k[x] \bmod g(x)$ is a domain iff $(g(x))$ is a prime ideal iff $g(x)$ is a prime element in $k[x]$ iff $g(x)$ is irreducible in $k[x]$. Combined with what we showed in Proposition 1. The results can even be strengthened to $k[x] \bmod g(x)$ is a domain iff $(g(x))$ is a prime ideal iff $g(x)$ is a prime element in $k[x]$ iff $g(x)$ is irreducible in $k[x]$ iff $k[x] \bmod g(x)$ is a field. Note that this is saying $k[x] \bmod g(x)$ is a domain iff it is a field!